

No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	
Project Name	Rising water
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Registration	Allows new users to securely create an account using hashed passwords.	Authentication	Done	
2	Secure User Login	Authenticates existing users and manages active sessions securely.	Authentication	Done	
3	Flood Prediction Engine	Processes real-time meteorological inputs to predict flood probability using XGBoost.	ML / Backend	Done	
4	Prediction History	Persistently stores and displays past user predictions via an interactive dashboard.	Database	Done	
5	Application Health API	A dedicated /health endpoint to monitor backend routing speed and system uptime.	API / Integration	Done	
6	Dynamic UI Templating	Responsive web forms dynamically rendered based	Frontend (UI)	Done	

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
		on user login state.			

Feature Summary:

Metric	Count / Value
Total Features Planned	6
Total Features Implemented	6
Core / Must-Have Features	4
Additional / Nice-to-Have Features	2
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Responsive HTML/CSS Web Forms, Dynamic Jinja2 Data Rendering
2	Backend / Logic	2	XGBoost Inference Execution, Flask Route Controllers (app.py)
3	Database / Storage	2	SQLAlchemy User Tracking, Persistent Prediction Logging
4	API / Integration	2	/predict Data Payload Endpoint, /health Uptime Endpoint
5	Security / Authentication	2	Werkzeug Password Hashing, Flask-Login Session Management